Problem Sheet – 2 Solutions: Improper Integrals

1. Evaluate the following improper integrals.

(a)

$$I = \int_{-1}^{1} \frac{dx}{x^{2/3}}.$$

The integrand is improper at $x = 0$. Therefore,

$$I = \int_{-1}^{0} \frac{dx}{x^{2/3}} + \int_{0}^{1} \frac{dx}{x^{2/3}}.$$

Since

$$\frac{1}{x^{2/3}} = |x|^{-2/3},$$

both integrals are equal. Moreover,

$$\int_0^1 x^{-2/3}\,dx = \lim_{\varepsilon\to 0^+} \int_\varepsilon^1 x^{-2/3}\,dx = \lim_{\varepsilon\to 0^+} \left[3x^{1/3}\right]_\varepsilon^1 = 3.$$

Hence

$$\boxed{I = 6}.$$

(b)

$$I = \int_{-8}^{1} \frac{dx}{x^{1/3}}.$$

The integrand is improper at $x = 0$. Thus

$$I = \int_{-8}^{0} \frac{dx}{x^{1/3}} + \int_{0}^{1} \frac{dx}{x^{1/3}}.$$

Since

$$\int x^{-1/3}\,dx = \frac{3}{2}x^{2/3},$$

we obtain

$$\int_{-8}^{0} \frac{dx}{x^{1/3}} = \lim_{\varepsilon\to 0^+} \frac{3}{2}\left((-\varepsilon)^{2/3} - (-8)^{2/3}\right) = -\frac{3}{2}(4) = -6.$$

Also,

$$\int_0^1 x^{-1/3}\,dx = \frac{3}{2}.$$

Therefore,

$$\boxed{I = -\frac{9}{2}}.$$

(c)

$$I = \int_0^1 \frac{dx}{\sqrt{1-x^2}}.$$

The integrand is improper at $x = 1$. We have

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x.$$

Therefore,

$$I = \lim_{b\to 1^-} \left[\sin^{-1} x\right]_0^b = \sin^{-1}(1) - \sin^{-1}(0) = \frac{\pi}{2}.$$

Hence

$$\boxed{I = \frac{\pi}{2}}.$$

(d)

$$I = \int_0^1 \frac{dr}{r^{0.999}}.$$

Since

$$0.999 = \frac{999}{1000},$$

we have

$$I = \int_0^1 r^{-0.999}\, dr.$$

Using

$$\int r^{-0.999}\, dr = \frac{r^{0.001}}{0.001} = 1000 r^{0.001},$$

we get

$$I = \lim_{\varepsilon\to 0^+} 1000\left(1 - \varepsilon^{0.001}\right) = 1000.$$

Thus

$$\boxed{I = 1000}.$$

(e)

$$I = \int_{-\infty}^{-2} \frac{2\, dx}{x^2 - 1}.$$

We use

$$\frac{2}{x^2-1} = \frac{1}{x-1} - \frac{1}{x+1}.$$

Hence

$$\int \frac{2\, dx}{x^2-1} = \ln\left|\frac{x-1}{x+1}\right|.$$

Therefore,

$$\begin{aligned} I &= \lim_{a\to-\infty}\left[\ln\left|\frac{x-1}{x+1}\right|\right]_a^{-2} \\ &= \ln\left|\frac{-3}{-1}\right| - \lim_{a\to-\infty}\ln\left|\frac{a-1}{a+1}\right| \\ &= \ln 3 - \ln 1. \end{aligned}$$

Thus

$$\boxed{I = \ln 3}.$$

(f)

$$I = \int_{-\infty}^{2} \frac{2\,dx}{x^2+4}.$$

Since

$$\int \frac{2\,dx}{x^2+4} = \tan^{-1}\left(\frac{x}{2}\right),$$

we obtain

$$\begin{aligned} I &= \lim_{a\to-\infty}\left[\tan^{-1}\left(\frac{x}{2}\right)\right]_a^2 \\ &= \frac{\pi}{4} - \left(-\frac{\pi}{2}\right) = \frac{3\pi}{4}. \end{aligned}$$

Hence

$$\boxed{I = \frac{3\pi}{4}}.$$

(g)

$$I = \int_0^{\infty} \frac{dx}{(1+x)\sqrt{x}}.$$

Put

$$x = t^2, \qquad dx = 2t\,dt, \qquad \sqrt{x} = t.$$

Then

$$I = \int_0^{\infty} \frac{2t}{(1+t^2)t}\,dt = 2\int_0^{\infty} \frac{dt}{1+t^2}.$$

Therefore,

$$I = 2\left[\tan^{-1} t\right]_0^{\infty} = 2\cdot\frac{\pi}{2} = \boxed{\pi}.$$

(h)

$$I = \int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}}.$$

Use the substitution

$$x = \sec\theta, \qquad dx = \sec\theta\tan\theta\,d\theta, \qquad \sqrt{x^2-1} = \tan\theta.$$

When $x = 1$, $\theta = 0$, and when $x \to \infty$, $\theta \to \frac{\pi}{2}$.
Hence

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\sec\theta \tan\theta}{\sec\theta \tan\theta}\, d\theta \\ &= \int_0^{\pi/2} d\theta = \boxed{\frac{\pi}{2}}. \end{aligned}$$

(i)

$$I = \int_0^{\infty} \frac{dv}{(1+v^2)(1+\tan^{-1} v)}.$$

Let

$$u = \tan^{-1} v.$$

Then

$$du = \frac{dv}{1+v^2}.$$

Also,

$$v = 0 \implies u = 0, \qquad v \to \infty \implies u \to \frac{\pi}{2}.$$

Thus

$$I = \int_0^{\pi/2} \frac{du}{1+u} = \left[\ln(1+u)\right]_0^{\pi/2}.$$

Therefore,

$$\boxed{I = \ln\left(1 + \frac{\pi}{2}\right)}.$$

(j)

$$I = \int_0^{\infty} \frac{16\tan^{-1} x}{1+x^2}\, dx.$$

Put

$$u = \tan^{-1} x, \qquad du = \frac{dx}{1+x^2}.$$

As x goes from 0 to ∞, u goes from 0 to $\frac{\pi}{2}$. Hence

$$\begin{aligned} I &= 16 \int_0^{\pi/2} u\, du \\ &= 16 \left[\frac{u^2}{2}\right]_0^{\pi/2} \\ &= 8\frac{\pi^2}{4}. \end{aligned}$$

Therefore,

$$\boxed{I = 2\pi^2}.$$

(k)

$$I = \int_{-\infty}^{0} \theta e^{\theta}\, d\theta.$$

Integration by parts gives

$$\int \theta e^{\theta}\, d\theta = e^{\theta}(\theta - 1).$$

Therefore,

$$\begin{aligned} I &= \lim_{a \to -\infty} \left[e^{\theta}(\theta - 1) \right]_{a}^{0} \\ &= -1 - \lim_{a \to -\infty} e^{a}(a - 1). \end{aligned}$$

Since

$$\lim_{a \to -\infty} e^{a}(a - 1) = 0,$$

we obtain

$$\boxed{I = -1}.$$

(l)

$$I = \int_{0}^{\infty} 2e^{-\theta} \sin\theta\, d\theta.$$

We use the standard formula

$$\int e^{a\theta} \sin(b\theta)\, d\theta = \frac{e^{a\theta}}{a^2 + b^2} \left(a \sin(b\theta) - b \cos(b\theta) \right).$$

With $a = -1$ and $b = 1$,

$$\int 2e^{-\theta} \sin\theta\, d\theta = -e^{-\theta}(\sin\theta + \cos\theta).$$

Hence

$$\begin{aligned} I &= \lim_{b \to \infty} \left[-e^{-\theta}(\sin\theta + \cos\theta) \right]_{0}^{b} \\ &= 0 - (-1) = 1. \end{aligned}$$

Thus

$$\boxed{I = 1}.$$

(m)

$$I = \int_{-\infty}^{0} e^{-|x|}\, dx.$$

For $x < 0$,

$$|x| = -x,$$

and hence

$$e^{-|x|} = e^{x}.$$

Therefore,

$$I = \int_{-\infty}^{0} e^x \, dx = [e^x]_{-\infty}^{0} = 1.$$

Thus

$$\boxed{I = 1}.$$

(n)

$$I = \int_{-\infty}^{\infty} 2xe^{-x^2} \, dx.$$

First observe that

$$\int_{0}^{\infty} 2xe^{-x^2} \, dx = \left[-e^{-x^2}\right]_{0}^{\infty} = 1,$$

while

$$\int_{-\infty}^{0} 2xe^{-x^2} \, dx = -1.$$

Both one-sided improper integrals converge. Therefore, by the definition of an improper integral over the whole real line,

$$I = \int_{-\infty}^{0} 2xe^{-x^2} \, dx + \int_{0}^{\infty} 2xe^{-x^2} \, dx = -1 + 1.$$

Hence

$$\boxed{I = 0}.$$

2. Discuss the convergence of the following improper integrals.

(a)

$$I = \int_{0}^{1} \frac{\ln x}{x^2} \, dx.$$

The integrand is improper at $x = 0$. Integration by parts gives

$$\int \frac{\ln x}{x^2} \, dx = -\frac{\ln x + 1}{x}.$$

Therefore,

$$\begin{aligned} I &= \lim_{\varepsilon \to 0^+} \left[-\frac{\ln x + 1}{x} \right]_{\varepsilon}^{1} \\ &= -1 + \lim_{\varepsilon \to 0^+} \frac{\ln \varepsilon + 1}{\varepsilon}. \end{aligned}$$

Since

$$\lim_{\varepsilon \to 0^+} \frac{\ln \varepsilon + 1}{\varepsilon} = -\infty,$$

we conclude that

$$\boxed{\int_0^1 \frac{\ln x}{x^2}\,dx = -\infty.}$$

Thus the improper integral **diverges**.

(b)

$$I = \int_1^2 \frac{dx}{x \ln x}.$$

The integrand is improper at $x = 1$. Since

$$\int \frac{dx}{x \ln x} = \ln|\ln x|,$$

we have

$$\begin{aligned} I &= \lim_{\varepsilon \to 0^+} [\ln(\ln x)]_{1+\varepsilon}^2 \\ &= \ln(\ln 2) - \lim_{\varepsilon \to 0^+} \ln(\ln(1+\varepsilon)). \end{aligned}$$

Since

$$\ln(1+\varepsilon) \to 0^+$$

and hence

$$\ln(\ln(1+\varepsilon)) \to -\infty,$$

we obtain

$$\boxed{I = +\infty}.$$

Thus the improper integral **diverges**.

(c)

$$I = \int_0^{\ln 2} x^{-2} e^{-1/x}\,dx.$$

Put

$$u = \frac{1}{x}.$$

Then

$$dx = -\frac{1}{u^2}\,du, \qquad x^{-2} = u^2.$$

As $x \to 0^+$, $u \to \infty$, while

$$x = \ln 2 \quad \Longrightarrow \quad u = \frac{1}{\ln 2}.$$

Thus

$$\begin{aligned} I &= \int_\infty^{1/\ln 2} u^2 e^{-u} \left(-\frac{1}{u^2}\right) du \\ &= \int_{1/\ln 2}^\infty e^{-u}\,du \\ &= \left[-e^{-u}\right]_{1/\ln 2}^\infty. \end{aligned}$$

Therefore,

$$\boxed{I = e^{-1/\ln 2}}.$$

(d)

$$I = \int_0^1 \frac{e^{-\sqrt{x}}}{\sqrt{x}}\, dx.$$

Put

$$u = \sqrt{x}.$$

Then

$$x = u^2, \qquad dx = 2u\, du.$$

Therefore,

$$I = \int_0^1 \frac{e^{-u}}{u}(2u)\, du = 2\int_0^1 e^{-u}\, du.$$

Hence

$$I = 2[-e^{-u}]_0^1 = 2(1 - e^{-1}).$$

Thus

$$\boxed{I = 2(1 - e^{-1})}.$$

(e)

$$I = \int_1^\infty \frac{dx}{x^3 + 1}.$$

Factor

$$x^3 + 1 = (x + 1)(x^2 - x + 1).$$

Partial fractions give

$$\frac{1}{x^3 + 1} = \frac{1}{3(x + 1)} + \frac{-x + 2}{3(x^2 - x + 1)}.$$

An antiderivative is

$$\int \frac{dx}{x^3 + 1} = \frac{1}{3}\ln(x + 1) - \frac{1}{6}\ln(x^2 - x + 1) + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x - 1}{\sqrt{3}}\right).$$

Consequently,

$$\begin{aligned} I = \lim_{b\to\infty} &\left[\frac{1}{3}\ln(b + 1) - \frac{1}{6}\ln(b^2 - b + 1) + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2b - 1}{\sqrt{3}}\right)\right] \\ &- \left[\frac{1}{3}\ln 2 - \frac{1}{6}\ln 1 + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)\right]. \end{aligned}$$

Since

$$\frac{1}{3}\ln(b + 1) - \frac{1}{6}\ln(b^2 - b + 1) \longrightarrow 0$$

and

$$\tan^{-1}\left(\frac{2b-1}{\sqrt{3}}\right) \longrightarrow \frac{\pi}{2},$$

we obtain

$$I = \frac{\pi}{2\sqrt{3}} - \frac{1}{3}\ln 2 - \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6}.$$

Thus

$$\boxed{I = \frac{\pi}{3\sqrt{3}} - \frac{1}{3}\ln 2}.$$

(f)

$$I = \int_4^\infty \frac{dx}{\sqrt{x}-1}.$$

For large x,

$$\sqrt{x} - 1 \leq \sqrt{x},$$

and hence

$$\frac{1}{\sqrt{x}-1} \geq \frac{1}{\sqrt{x}} = \frac{1}{x^{1/2}}.$$

But

$$\int_4^\infty \frac{dx}{\sqrt{x}} = \int_4^\infty x^{-1/2}\,dx$$

diverges.
Therefore, by the comparison test,

$$\boxed{\int_4^\infty \frac{dx}{\sqrt{x}-1} \text{ diverges to } +\infty.}$$

(g)

$$I = \int_2^\infty \frac{dv}{\sqrt{v-1}}.$$

Put

$$u = v - 1.$$

Then $du = dv$, and

$$I = \int_1^\infty u^{-1/2}\,du.$$

Thus

$$I = \lim_{b\to\infty} \left[2\sqrt{u}\right]_1^{b-1} = \lim_{b\to\infty} 2\left(\sqrt{b-1} - 1\right) = \infty.$$

Therefore,

$$\boxed{\text{the integral diverges to } +\infty.}$$

(h)

$$I = \int_0^\infty \frac{d\theta}{1+e^\theta}.$$

We use

$$\frac{1}{1+e^\theta} = \frac{e^{-\theta}}{1+e^{-\theta}}.$$

Alternatively, put

$$u = e^{-\theta}.$$

Then

$$du = -e^{-\theta}\,d\theta, \qquad d\theta = -\frac{du}{u},$$

and

$$\frac{1}{1+e^\theta} = \frac{u}{1+u}.$$

As θ goes from 0 to ∞, u goes from 1 to 0. Hence

$$\begin{aligned} I &= \int_1^0 \frac{u}{1+u}\left(-\frac{du}{u}\right) \\ &= \int_0^1 \frac{du}{1+u} \\ &= [\ln(1+u)]_0^1 \\ &= \ln 2. \end{aligned}$$

Thus

$$\boxed{I = \ln 2}.$$

(i)

$$I = \int_\pi^\infty \frac{2+\cos x}{x}\,dx.$$

Split the integral:

$$I = 2\int_\pi^\infty \frac{dx}{x} + \int_\pi^\infty \frac{\cos x}{x}\,dx.$$

The first integral diverges:

$$2\int_\pi^\infty \frac{dx}{x} = +\infty.$$

On the other hand,

$$\int_\pi^\infty \frac{\cos x}{x}\,dx$$

converges by Dirichlet's test, since $1/x$ decreases monotonically to 0 and the partial integrals of $\cos x$ are bounded.
Therefore, the divergent logarithmic term cannot be cancelled by the convergent oscillatory term. Hence

$$\boxed{\int_{\pi}^{\infty} \frac{2+\cos x}{x}\,dx \text{ diverges to } +\infty.}$$

(j)

$$I = \int_{\pi}^{\infty} \frac{1+\sin x}{x^2}\,dx.$$

Since

$$-1 \le \sin x \le 1,$$

we have

$$0 \le \frac{1+\sin x}{x^2} \le \frac{2}{x^2}.$$

The integral

$$\int_{\pi}^{\infty} \frac{2}{x^2}\,dx$$

converges. Therefore, by the comparison test,

$$\boxed{\int_{\pi}^{\infty} \frac{1+\sin x}{x^2}\,dx \text{ converges.}}$$

In fact, it can be evaluated by integration by parts:

$$\int_{\pi}^{\infty} \frac{\sin x}{x^2}\,dx = \left[-\frac{\sin x}{x}\right]_{\pi}^{\infty} + \int_{\pi}^{\infty} \frac{\cos x}{x}\,dx = \int_{\pi}^{\infty} \frac{\cos x}{x}\,dx.$$

Thus

$$I = \frac{1}{\pi} + \int_{\pi}^{\infty} \frac{\cos x}{x}\,dx.$$

The latter integral is the cosine integral and does not have an elementary closed form. Hence the integral is convergent, with

$$\boxed{I = \frac{1}{\pi} + \int_{\pi}^{\infty} \frac{\cos x}{x}\,dx.}$$

3. Show that

$$\int_{0}^{\infty} \frac{2x}{x^2+1}\,dx$$

diverges and hence that

$$\int_{-\infty}^{\infty} \frac{2x}{x^2+1}\,dx$$

diverges. Then show that

$$\lim_{b\to\infty} \int_{-b}^{b} \frac{2x}{x^2+1}\,dx = 0.$$

Solution

First consider

$$\int_0^\infty \frac{2x}{x^2+1}\,dx.$$

Since

$$\int \frac{2x}{x^2+1}\,dx = \ln(x^2+1),$$

we have

$$\begin{aligned}\int_0^\infty \frac{2x}{x^2+1}\,dx &= \lim_{b\to\infty}\left[\ln(x^2+1)\right]_0^b \\ &= \lim_{b\to\infty}\ln(b^2+1) \\ &= +\infty.\end{aligned}$$

Therefore,

$$\boxed{\int_0^\infty \frac{2x}{x^2+1}\,dx \text{ diverges.}}$$

Now consider

$$\int_{-\infty}^\infty \frac{2x}{x^2+1}\,dx.$$

By the definition of an improper integral over the whole real line,

$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^0 f(x)\,dx + \int_0^\infty f(x)\,dx,$$

provided both one-sided integrals converge.
Here,

$$\int_0^\infty \frac{2x}{x^2+1}\,dx = +\infty.$$

Similarly,

$$\int_{-\infty}^0 \frac{2x}{x^2+1}\,dx = -\infty.$$

Indeed,

$$\begin{aligned}\int_{-\infty}^0 \frac{2x}{x^2+1}\,dx &= \lim_{a\to-\infty}\left[\ln(x^2+1)\right]_a^0 \\ &= -\lim_{a\to-\infty}\ln(a^2+1) \\ &= -\infty.\end{aligned}$$

Thus the improper integral

$$\boxed{\int_{-\infty}^\infty \frac{2x}{x^2+1}\,dx}$$

does not exist.
However, consider the symmetric integral

$$\int_{-b}^{b} \frac{2x}{x^2+1}\,dx.$$

The function

$$f(x) = \frac{2x}{x^2+1}$$

is odd. Therefore, for every $b > 0$,

$$\int_{-b}^{b} \frac{2x}{x^2+1}\,dx = 0.$$

Consequently,

$$\boxed{\lim_{b\to\infty} \int_{-b}^{b} \frac{2x}{x^2+1}\,dx = 0.}$$

Thus,

$$\boxed{\int_{-\infty}^{\infty} \frac{2x}{x^2+1}\,dx \text{ does not exist, whereas its Cauchy principal value is } 0.}$$

Indeed, the Cauchy principal value is defined by

$$\mathrm{PV} \int_{-\infty}^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_{-b}^{b} f(x)\,dx,$$

and therefore

$$\boxed{\mathrm{PV} \int_{-\infty}^{\infty} \frac{2x}{x^2+1}\,dx = 0.}$$

This demonstrates that, in general,

$$\boxed{\int_{-\infty}^{\infty} f(x)\,dx \neq \lim_{a\to\infty} \int_{-a}^{a} f(x)\,dx.}$$

The right-hand side, when it exists, is the *Cauchy principal value*; it is not necessarily the improper integral.